



BITS Pilani

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AIML for Robotics

Module 2 – Sensors, Measurement, Model

Session Faculty:

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The course owner and the authors of this course deck, Prof. Raja vadhana and Dr.Sangeetha, are gratefully acknowledging all authors who made their course materials freely available online.

☐ Introduction

- ☐ Types of Robots
- ☐ Mobile Robots – Use case of Programmable solutions

☐ Components of Programmable Robotic System

- ☐ Perception
- ☐ Representation
- ☐ Motion

☐ Localization

- ☐ Taxonomy
- ☐ Types

☐ Mapping

- ☐ Feature Based
- ☐ Grid Based

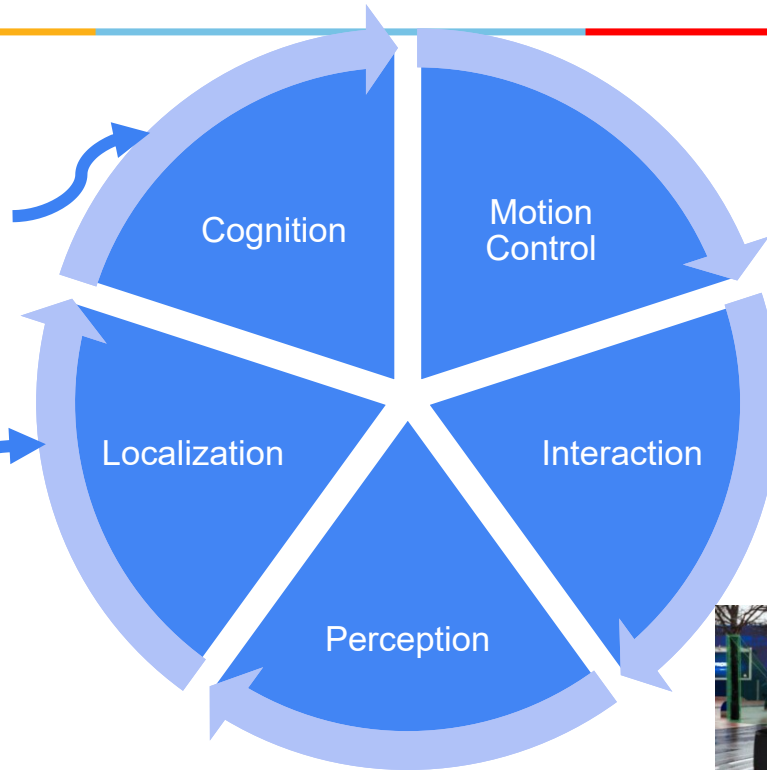
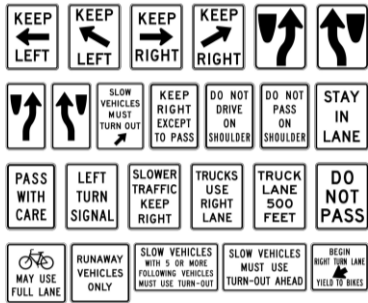
☐ Path Planning

- ☐ Graph Based
- ☐ Sample Based
- ☐ Evolutional Techniques

☐ Human – Robot Interaction

- ☐ Ethics
- ☐ Introduction to Visual Bot
- ☐ Introduction to Chat Bot

Perception & Control



x_0	$P(x_0)$
is_open	0.5
is_closed	0.5

x_{t-1}	u_t	$P(x_t = \text{is_open})$	$P(x_t = \text{is_closed})$
is_open	push	1	0
is_closed	Push	0.8	0.2
is_open	do_nothing	1	0
is_closed	do_nothing	0	1

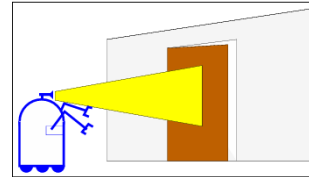
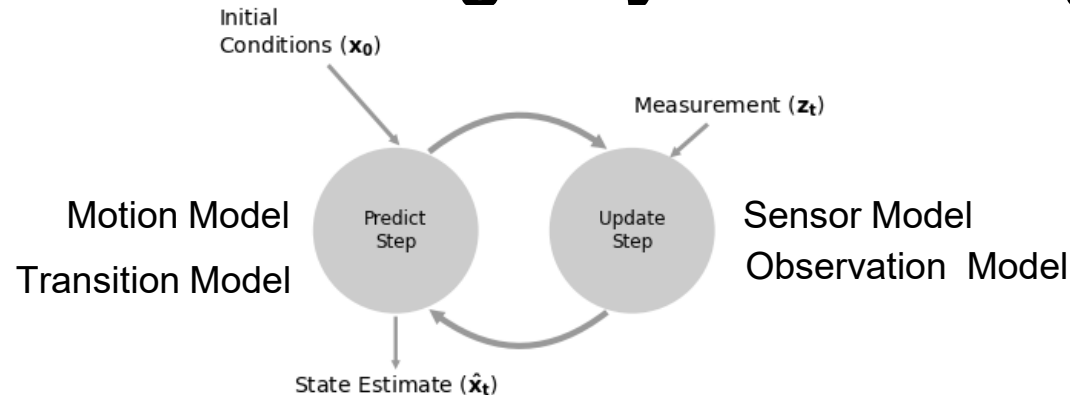


Figure 2.2 A mobile robot estimating the state of a door.

x_t	$P(z_t = \text{sense_open})$	$P(z_t = \text{sense_closed})$
is_open	0.6	0.4
is_closed	0.2	0.8

State Estimation using Bayes Filter Algorithm



Recap

1. Initial Belief
2. Control Update
3. Measurement Update

X_{t-1}	U_t	$P(X_t = \text{is_open})$	$P(X_t = \text{is_closed})$
is_open	push	1	0
is_closed	Push	0.8	0.2
is_open	do_nothing	1	0
is_closed	do_nothing	0	1

SENSOR MEASUREMENT MODEL : $P(Z_t | X_t)$

X_t	$P(Z_t = \text{sense_open})$	$P(Z_t = \text{sense_closed})$
is_open	0.6	0.4
is_closed	0.2	0.8

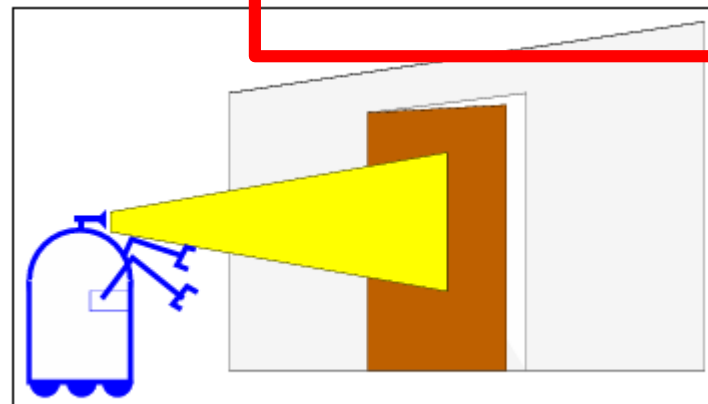
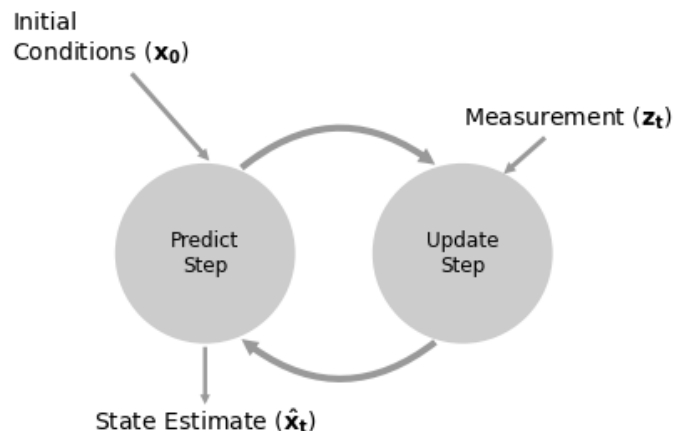
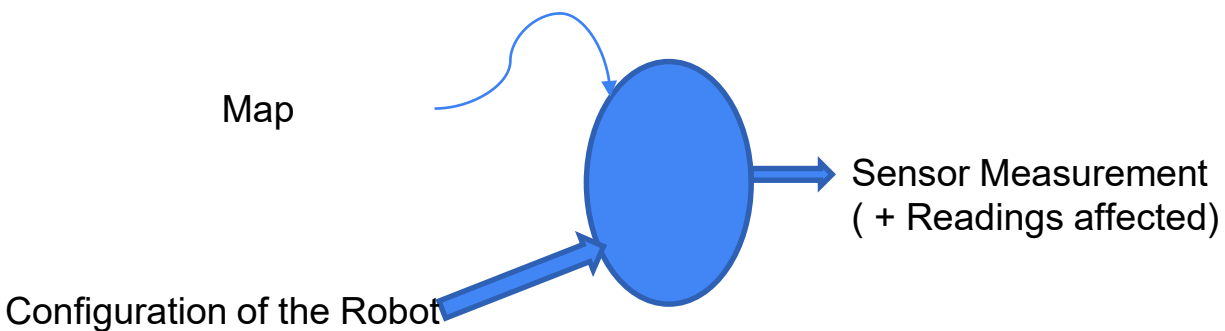


Figure 2.2 A mobile robot estimating the state of a door.

Inference :

The robot believes that with 0.983 probability the door is open

Sensor Model



Measurement Model

Why Must It Be Probabilistic?



CLASSICAL VIEW (DETERMINISTIC)

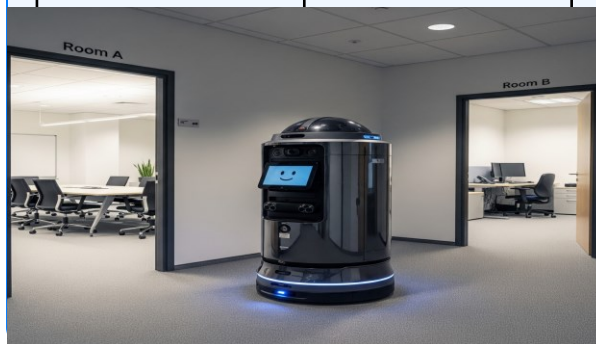
Assumption: Sensor Reading = Reality

Wall at 5m + Laser says 5m
= **Correct**

Failure Points:

- Sensor Noise & Drift
- Dynamic Obstacles
- Reflection Errors

Room A	0.6
Room B	0.3
Corridor	0.1



CYBERSECURITY CASE

Observations:

- Unusual login time
- Foreign IP
- Failed attempts

Probabilistic Conclusion:

$P(\text{attack} \mid \text{obs})$

"A robot never knows truth, only evidence."



Foundations of Probabilistic Sensing: $P(Z|X)$ vs $P(X|Z)$

1. MEASUREMENT MODEL $P(Z|X)$

"How likely is the sensor reading Z given state X ?"

Forward Sensor Model:

- Expected: $Z^* = 5\text{m}$
- Actual: $Z = 4.8\text{m}$

$P(Z=4.8 \mid X)$

Spam Analogy:

"How likely a spam email contains the word 'lottery'?"

BAYES RULE RELATIONSHIP

$$P(X|Z) = \frac{P(Z|X) P(X)}{P(Z)}$$

$P(Z|X)$: Sensor Physics

$P(X)$: Prior Belief

$P(X|Z)$: Updated Belief

sync

Foundations of Probabilistic Sensing: $P(Z|X)$ vs $P(X|Z)$

1. MEASUREMENT MODEL $P(Z|X)$

"How likely is the sensor reading Z given state X ?"

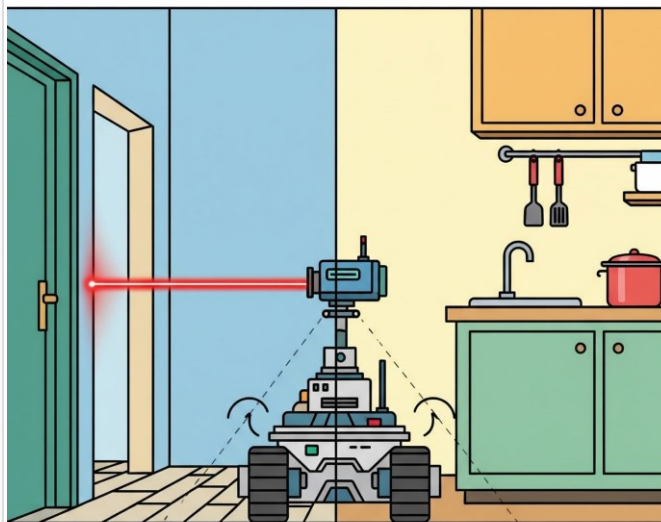
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"How likely a spam email contains the word 'lottery'?"



TOY NUMERICAL EXAMPLE

Priors:

$$P(\text{Hall}) = 0.7 \mid P(\text{Kit}) = 0.3$$

Likelihoods (Sound):

$$P(s|\text{Hall}) = 0.1 \mid P(s|\text{Kit}) = 0.9$$

Posterior Calculation:

$$\text{Numerator} = 0.9 * 0.3 = 0.27$$

$$\text{Normalize: } P(\text{sound}) = 0.34$$

$$P(\text{Kit}|\text{sound}) = 0.794$$

Result: Strong belief in Kitchen

Foundations of Probabilistic Sensing: $P(Z|X)$ vs $P(X|Z)$

1. MEASUREMENT MODEL $P(Z|X)$

"How likely is the sensor reading Z given state X ?"

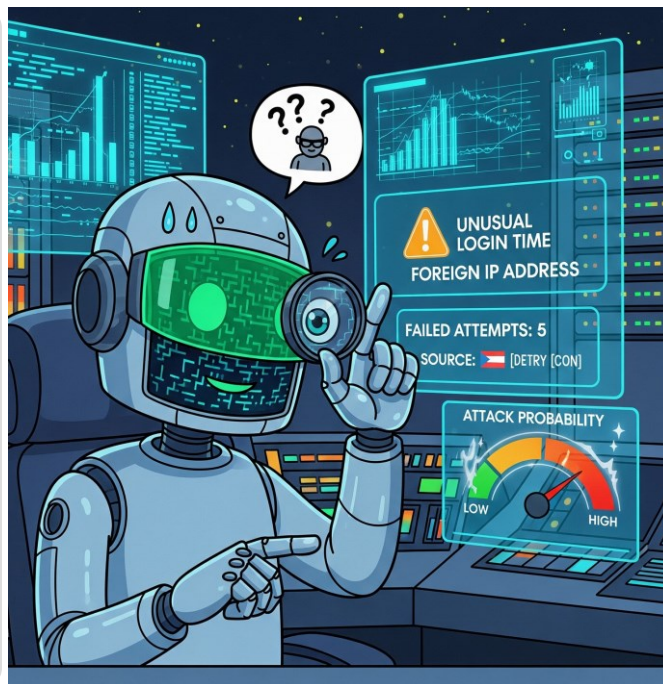
Forward Sensor Model:

- Expected: $Z^* = 5m$
- Actual: $Z = 4.8m$

$$P(Z=4.8 \mid X)$$

Spam Analogy:

"How likely a spam email contains the word 'lottery'?"



TOY NUMERICAL EXAMPLE

Priors:

$$P(\text{Attack}) = 0.7 \mid P(\text{Normal}) = 0.3$$

Likelihoods (LoginTime):

$$P(I|\text{Attack}) = 0.9 \mid P(I|\text{Normal}) = 0.3$$

Posterior Calculation:

$$\text{Numerator} = 0.9 * 0.3 = 0.27$$

$$\text{Normalize: } P(\text{LoginTime}) = 0.34$$

$$P(\text{Normal}|\text{LoginTime}) = 0.794$$

Result: Strong belief in Attack scenario

Foundations of Probabilistic Sensing: $P(Z|X)$ vs $P(X|Z)$

1. DISTANCE SENSORS

Principle: $d = v \times t / 2$

- Sonar & LiDAR (Time-of-Flight)
- Laser: 20ns @ speed of light

Numerical Example:

$$d = (3 \times 10^8 \times 20 \times 10^{-9}) / 2 = 3\text{m}$$

2. MULTI-BEAM EVIDENCE

$$Z_t = \{z_t^1, z_t^2, \dots, z_t^k\}$$

Each beam is an independent evidence contributor to global belief.

Agent Analogy:

User clicks = Beams; Combined clicks = User preference belief.

BEAM-BASED RANGE MODEL

Likelihood Formula:

$$P(Z_t | X_t, m) = \prod P(z_t^k | X_t, m)$$

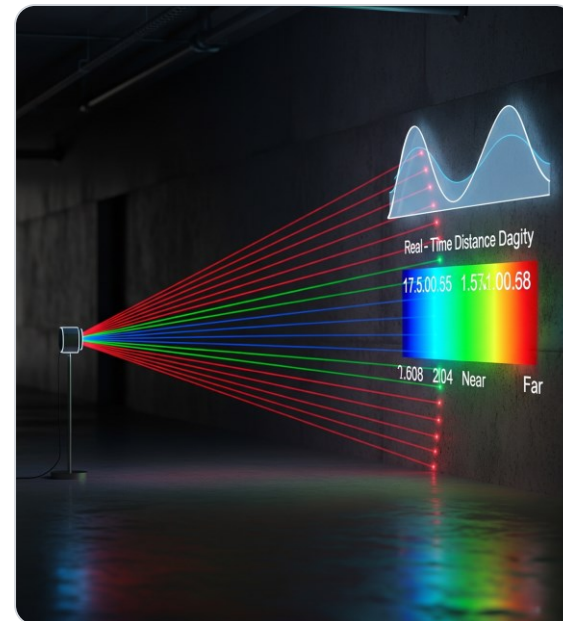
Case Study:

- Predict: Wall at 6m (Ray Trace)
- Measure: 5.8m (Actual Sensor)

$$P(z|x) = N(5.8; 6, 0.3^2)$$

Noise Approximation:

$$Z \sim N(5, 0.2^2)$$

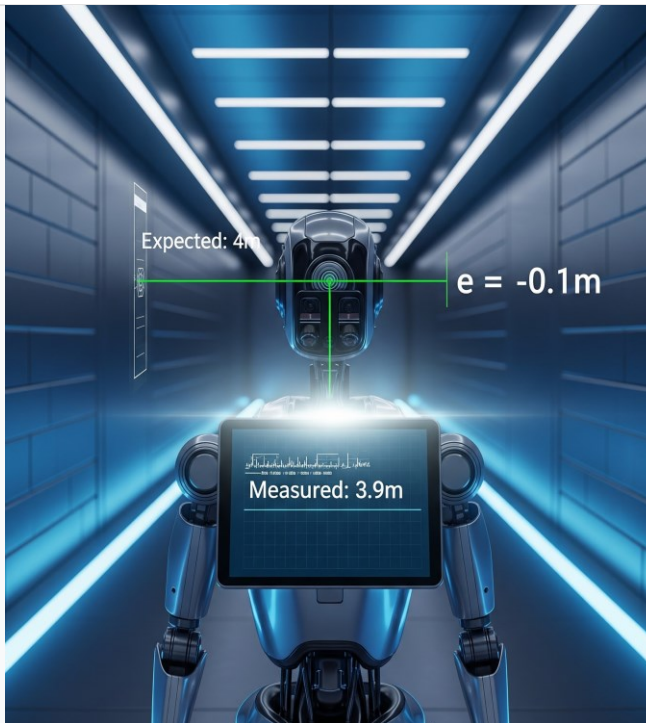


radar

Grounding Bayes' Rule: From Sound to Beam Models

THE SCENARIO

- **Hypothesis (X):** Robot is in the Hallway.
- **Map (m):** Hallway map predicts a wall at 4m (Z^* or Z_{expected}).
- **Measurement (Z):** Laser sensor reports 3.9m (Z_{measured}).



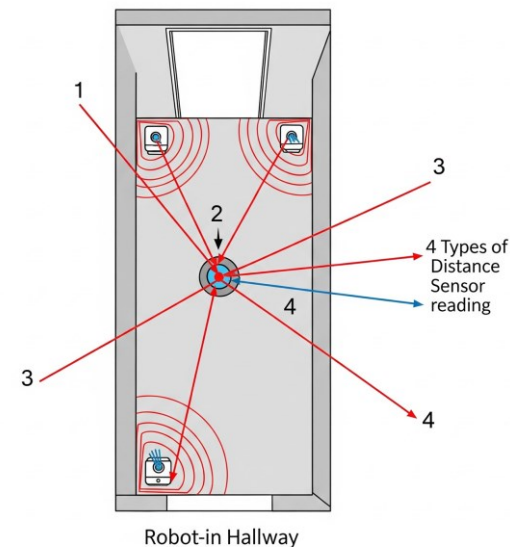
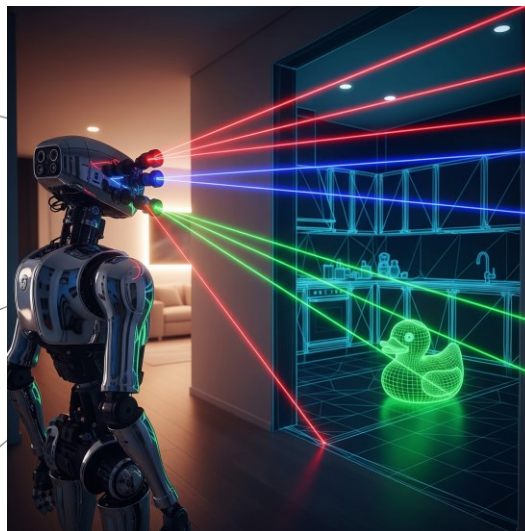
Model Idea!

1. Ray Casting
2. Error Calculation
3. Likelihood Computation

Beam-Based Measurement

WHY BEAM MODELS?

- Localization & Particle Filtering
- Occupancy Map Consistency
- Physics-based interpretability



Grounding Bayes' Rule: From Sound to Beam Models

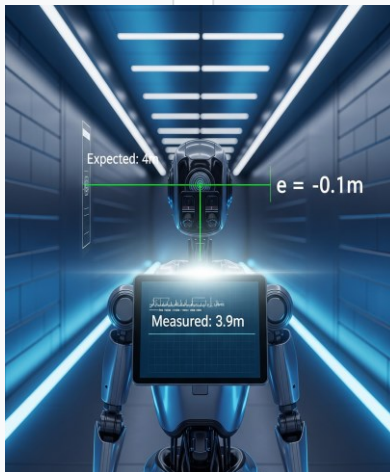
THE SCENARIO

Conceptually replace abstract likelihoods with values computed by a physical beam model.

- **Hypothesis (X):** Robot is in the Hallway.
- **Map (m):** Hallway map predicts a wall at 4m (Z^* or Z_{expected}).
- **Measurement (Z):** Laser sensor reports 3.9m (Z_{measured}).

INFERENCE

If $Z = 3.9\text{m}$ is the evidence, the computed likelihood $P(Z|\text{Hall}, m) \approx 0.882$ is the physically grounded value used as $P(Z|X)$ in Bayes' Rule.



LIKELIHOOD STEPS (BEAM MODEL)

1. Ray Casting

Determines expected measurement Z^* if truly in hallway. $Z^* = 4\text{m}$.

2. Error Calculation

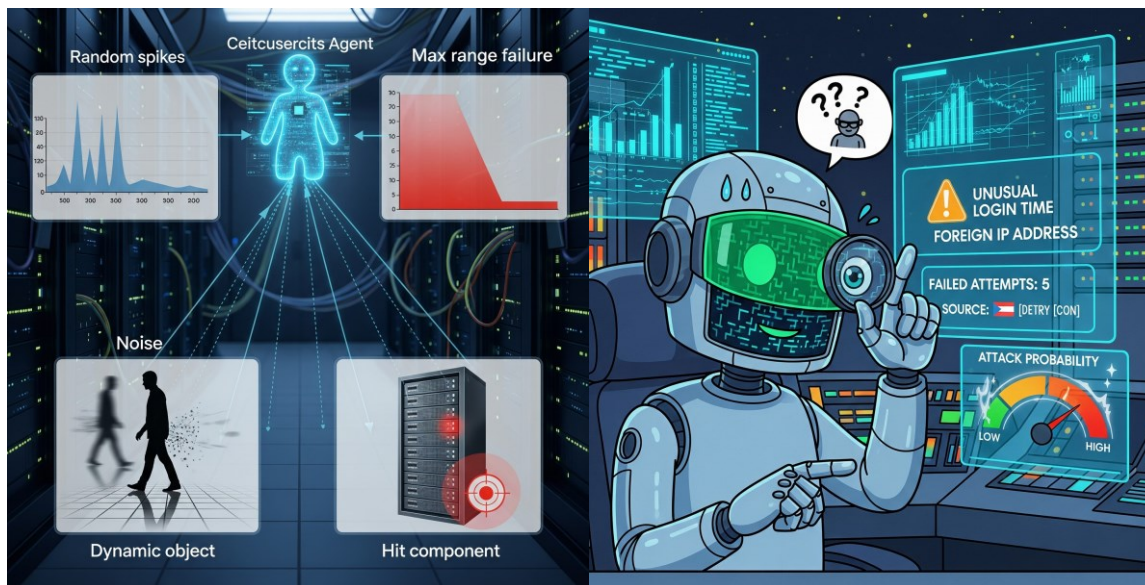
Difference between actual and predicted.
 $e = 3.9\text{m} - 4\text{m} = -0.1\text{m}$.

3. Likelihood Computation (Mixture Model)

$P(Z|\text{Hall}, m) = \text{whitPhit} + \text{wshortPshort} + \dots$

Gaussian Hit ($\sigma=0.2$, $e=-0.1$):
 $P \approx 0.882$ (High Likelihood)

Beam-Based Measurement



Beam-Based Measurement & Noise Models



1. BEAM LIKELIHOOD

Joint probability for K beams assuming independence:

$$P(Z|x, m) = \prod P(z|x, m)$$

Ray Casting: Simulated beam geometry predicts expected z^* based on map m

2. NUMERICAL EXAMPLE

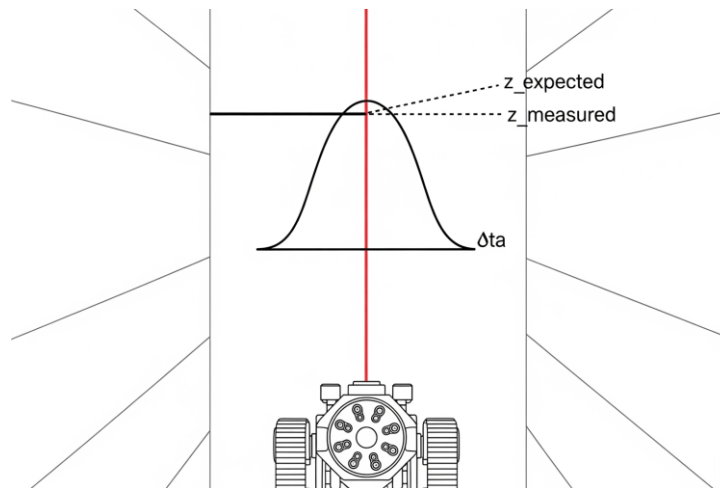
State: $x_t = (2, 2)$

Ray Trace: $z^* = 6.0\text{m}$

Actual: $z = 5.8\text{m}$

Gaussian Hit Likelihood:

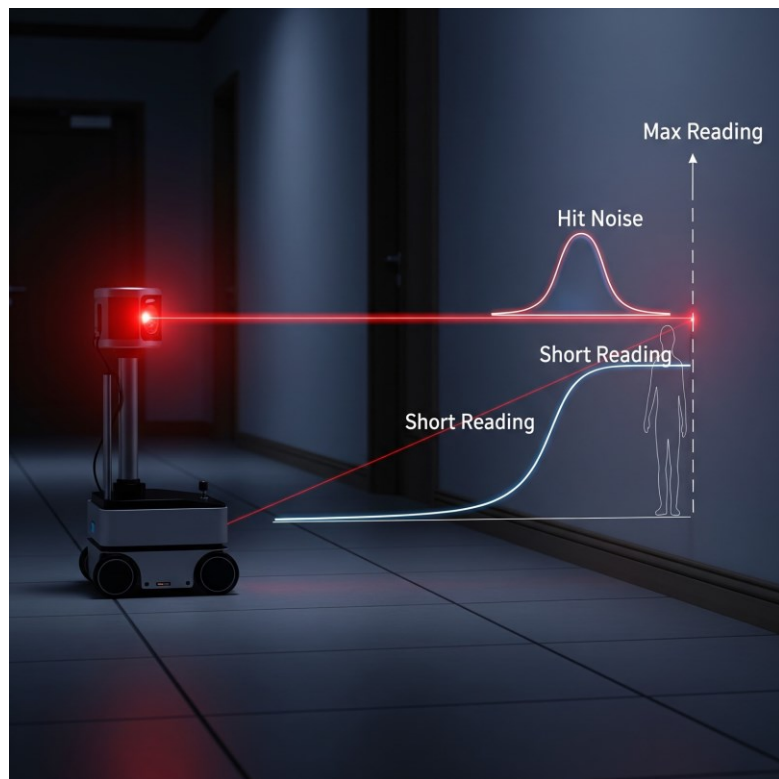
$$P(z|x)$$



WHY BEAM MODELS?

- Localization & Particle Filtering
- Occupancy Map Consistency
- Physics-based interpretability

Beam-Based Measurement & Noise Models



NOISE MIXTURE

Realistic sensing combines 4 types:

P_{hit} (Gaussian)

Normal sensor error.

P_{short} (Exponential)

Dynamic objects (humans).

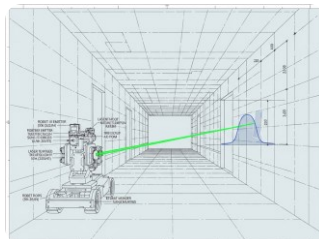
P_{max} (Delta)

Missed detection/failure.

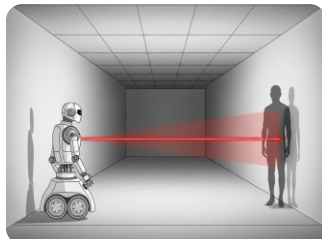
P_{rand} (Uniform)

Unexplainable noise.

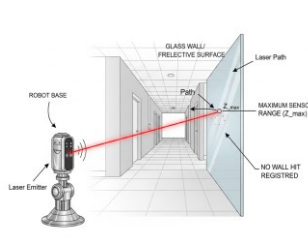
Beam-Based Measurement & Noise Models



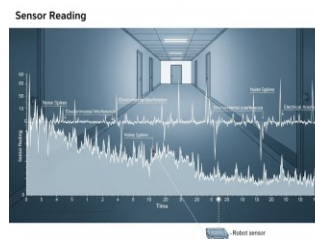
P_{hit}: Accuracy
Gaussian noise at wall



P_{short}: Dynamic
Blocked by human/object



P_{max}: Failure
Detection missed / Max



P_{rand}: Random
Electrical spikes/static

Hallway Numerical Insight

- **Scenario:** Robot at $X = 6\text{m}$, Wall at 10m . $Z_{\text{expected}} = 4\text{m}$.
- **Hit Example:** Measured $Z = 3.9\text{m}$. Error = -0.1m . High likelihood via Gaussian $N(0, \sigma^2)$.
- **Short Example:** Measured $Z = 1.0\text{m}$. Likely human at 1m . Modeled by Exponential $\lambda e^{-\lambda z}$.
- **Result:** Probabilistic model $P(Z|X, m)$ handles these ambiguities by mixing distributions.

NOISE MIXTURE

Realistic sensing combines 4 types:

P_{hit} (Gaussian)
Normal sensor error.

P_{short} (Exponential)
Dynamic objects (humans).

P_{max} (Delta)
Missed detection/failure.

P_{rand} (Uniform)
Unexplainable noise.

Beam-Based Measurement & Noise Models



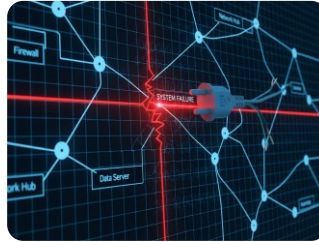
P_hit: Valid

Legitimate user access



P_short: Anomaly

Unexpected login attempt



P_max: Timeout

Server down / No response



P_rand: Jitter

Network packet noise

NOISE MIXTURE

Realistic sensing combines 4 types:

P_hit (Gaussian)

Normal sensor error.

P_short (Exponential)

Dynamic objects (humans).

P_max (Delta)

Missed detection/failure.

P_rand (Uniform)

Unexplainable noise.

Cybersecurity Numerical Insight

- **Scenario:** User usually logs in at **9 AM (X)**. **Expected Z = 08:00 to 10:00.**
- **Hit Example:** Login at **09:15 AM**. Error = +15m. High likelihood via Gaussian $N(0, \sigma^2)$.
- **Short Example:** Login at **03:00 AM**. Likely bot/attacker. Modeled by Exponential $\lambda e^{-\lambda z}$.
- **Result:** Probabilistic model $P(Z|X)$ handles these anomalies by mixing distributions.

Mixture Models: Sensor Noise Modeling



Complete Measurement Model

$$P(z|x) = w_{hit}P_{hit} + w_{short}P_{short} + w_{max}P_{max} + w_{rand}P_{rand}$$

Constraint: $\sum w_i = 1$

Component Physical Meaning

hit:

Normal detection (Gaussian noise)

short:

Unexpected object (e.g., moving people)

max:

Detection failure (max range)

rand:

Unexplained/Random reading

Why Mixture Models?

Handles real-world dynamics: moving people, reflections, transparent surfaces, and sensor noise.

Numerical Worked Example

Scenario: $z^* = 5\text{m}$ (Exp), $z = 4.7\text{m}$ (Obs)

Weights: $w_h:0.7$, $w_s:0.1$, $w_m:0.1$, $w_r:0.1$

- **Hit ($P_h=0.85$):** $0.7 \times 0.85 = 0.595$
- **Short ($P_s=0.3$):** $0.1 \times 0.3 = 0.030$
- **Max ($P_m=0$):** $0.1 \times 0 = 0.000$
- **Rand ($P_r=0.05$):** $0.1 \times 0.05 = 0.005$

Final $P(z|x) = 0.63$

Interpretation: Measurement strongly agrees with pose hypothesis.

Intrinsic Parameters: σ , λ , w_i

Methods: EM Algorithm, MLE, Bayesian Estimation 22

Deep Understanding of Measurement Models: $P(Z|X,m)$

CORE PROCESS: FROM SENSOR TO PROBABILITY

Finding $P(Z|X,m)$ is the process of evaluating how well an actual sensor reading (Z) matches what the robot expected to see (Z^*), given its assumed pose (X) and map (m).

- **Step 1 (Pose):** Robot assumes it is at location X (e.g., $x=6m$).
- **Step 2 (Map):** Map shows a wall at 10m.
- **Step 3 (Predict):** Expected $Z^* = 10 - 6 = 4m$.
- **Step 4 (Measure):** Sensor actually returns $Z = 3.8m$.
- **Step 5 (Evaluate):** Error $e = 3.8 - 4 = -0.2m$.

Is $P(Z|X,m)$ the Noise Distribution?

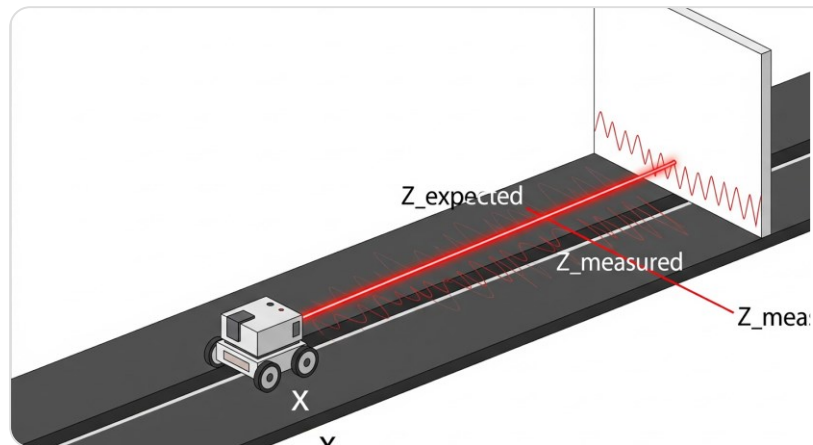
Not exactly. While the noise model defines the *shape* of the likelihood, the measurement model is the *application* of that noise relative to the map's geometry.

NUMERICAL EXAMPLE (GAUSSIAN HIT)

If sensor noise is Gaussian ($\sigma=0.2$) and error $e = -0.2$:

$$P(Z|X,m) = \eta * \exp(-(-0.2)^2 / (2 * 0.2^2)) = \eta * 0.606$$

A result of ~ 0.606 indicates a **high likelihood** that the robot is indeed at the hypothesized location.



Beam Model : Summary



CORE BEAM MODEL INTEGRATION

The Beam Model provides the physical grounding for the measurement model $P(\mathbf{Z}|\mathbf{X}, m)$ by simulating how light or sound interacts with the environment.

- **Physics-Based Interpretation:** Directly models sensor beams hitting obstacles in map m .
- **Joint Probability:** Combines K independent beams: $P(\mathbf{Z}|\mathbf{X}, m) = \prod P(z_k|\mathbf{X}, m)$.
- **Mixture Handling:** Integrates hit, short, max, and random noise types into one unified likelihood.

Required Reading

- T1 Chapter #5, #6
- T2 Chapter #5, #6.1, #6.2

Next Class Plan

- ✓ Measurement Model – Part 2
- ✓ AI-ML Techniques in Measurement Model